

Farhad Abedinzadeh

Postgraduate Researcher (PhD) in Bioengineering | Durham University

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Research profile

I am a postgraduate researcher (PhD) in Bioengineering at Durham University working on interpretable AI for early prediction of childhood myopia using longitudinal multimodal retinal imaging and biometric data. My research interests include OCT/OCTA analysis, retinal biomarkers, ocular biometry, longitudinal modelling and multimodal learning.

My previous research spans biomedical image and signal processing, computational neuroscience and machine learning, including EEG connectivity, EMG, gait, voice and MRI analysis in neurodevelopmental and neurodegenerative disorders.

Education

Current

PhD in Bioengineering · Durham University, United Kingdom

Research: Interpretable AI for Early Prediction of Childhood Myopia Using Longitudinal Multimodal Retinal Imaging and Biometric Data.

20 Jan 2021 - 11 Sep 2023

MSc in Sports Engineering

Islamic Azad University, Mashhad Branch, Iran · GPA: 17.40/20

Thesis: Detection of ADHD Using Phase-Based Brain Connectivity and Graph Theory.

23 Sep 2015 - 11 Nov 2020

BSc in Biomedical Engineering (Biomaterials)

Sadjad University of Technology, Mashhad, Iran

Thesis: Brain Tumor Detection Using Image Processing and Artificial Neural Networks.

Teaching & research experience

Durham

AI and Deep Learning · Python teaching materials

Preparation of practical notebooks and learning materials for students with limited Python experience.

May 2024

Instructor · EEG Signal Processing Workshop

Workshop held by the National Brain Mapping Laboratory in Mashhad, Iran.

Jul-Oct 2022

Instructor · Biomedical Image & Signal Processing Using MATLAB

Islamic Azad University, Mashhad Branch. Designed lectures, supervised practical sessions and guided projects in signal analysis, wavelets, pattern recognition and machine learning.

Joined Oct 2021

Research Assistant & Lab Manager

Research with Seyyed Abed Hosseini, Islamic Azad University, Mashhad Branch. Work on brain connectivity and multimodal data in neurological disorders.

Started Jan 2022

Biomedical programming & research project support

Python and MATLAB applications, data analysis, research methodology and support for undergraduate and master's research projects.

Joined Feb 2024

Venture Capital Analyst of Medical Devices

Razavi HighTech Industries, Mashhad, Iran.

Teaching development

Completed at Durham University.

- DCAD Teaching Induction: Teaching Skills: seminars and tutorials
- DCAD Teaching Induction: Assessment Skills
- DCAD Teaching Induction: Teaching Skills: lab teaching and science tutorials

Technical skills

Programming & machine learning

Python, MATLAB, PyTorch, TensorFlow/Keras, scikit-learn and Git.

Biomedical analysis

Image and signal preprocessing; EEG, EMG, ECG, gait and MRI analysis; connectivity, graph theory and time-frequency methods.

Retinal imaging research

OCT/OCTA exploration, EyePy workflows, retinal segmentation, thickness maps, en face projections and biometric integration.

Research tools

MNE, EEGLAB, Brainstorm, FieldTrip, BrainNet Viewer, BRAPH, HERMES, SPM, LaTeX, Mendeley, EndNote and SPSS.

Selected publications

Enhancing Multi-class Diabetic Retinopathy Detection Using Tuned Hyper-parameters and Modified Deep Transfer Learning

Journal article *Multimedia Tools and Applications* · 2024

Yeganeh Modaresnia, **Farhad Abedinzadeh**, Seyyed Abed Hosseini.

Improved ADHD Diagnosis Using EEG Connectivity and Deep Learning through Combining Pearson Correlation Coefficient and Phase-Locking Value

Journal article *Neuroinformatics* · 2024

Elham Ahmadi Moghadam, **Farhad Abedinzadeh**, Seyyed Abed Hosseini, Mohammad Hossein Moattar.

Simultaneous Time-Frequency Analysis of Gait Signals of Both Legs in Classifying Neurodegenerative Diseases

Journal article *Gait & Posture* · 2024

Farhad Abedinzadeh, Elham Ahmadi Moghadam, Seyyed Abed Hosseini.

Hybrid Deep Transfer Learning Based Early Diagnosis of Autism Spectrum Disorder Using Scalogram Representation of Electroencephalography Signals

Journal article *Medical & Biological Engineering & Computing* · 2024

Farhad Abedinzadeh, Yeganeh Modaresnia, Mohammad Hossein Moattar.

Complete bibliography: <https://farhadabedinzadeh.github.io/publications/>

Academic service

Journal peer reviewer:

- Iran Journal of Computer Science · Springer · Nov 2025-Present
- Multimedia Tools and Applications · Springer · Jan 2025-Present
- BMC Public Health · Springer · Dec 2024-Present
- Physical and Engineering Sciences in Medicine · Springer · Jan 2024-Present
- Computers in Biology and Medicine · Elsevier · Sep 2023-Present
- Pattern Recognition Letters · Elsevier · Jun 2023-Present
- Biomedical Signal Processing and Control · Elsevier · Feb 2023-Present
- Cognitive Neurodynamics
- Scientific Reports
- Discover Artificial Intelligence
- Signal, Image and Video Processing

Recognition

- Ranked second among Biomedical Engineering master's students at Islamic Azad University, Mashhad Branch.
- Finalist among 100 teams in the One Thousand Technological Ideas Tournament with an intelligent ring concept for HRV-based epileptic seizure prediction.

Languages

Persian: native · English: advanced (C1, IELTS Academic).